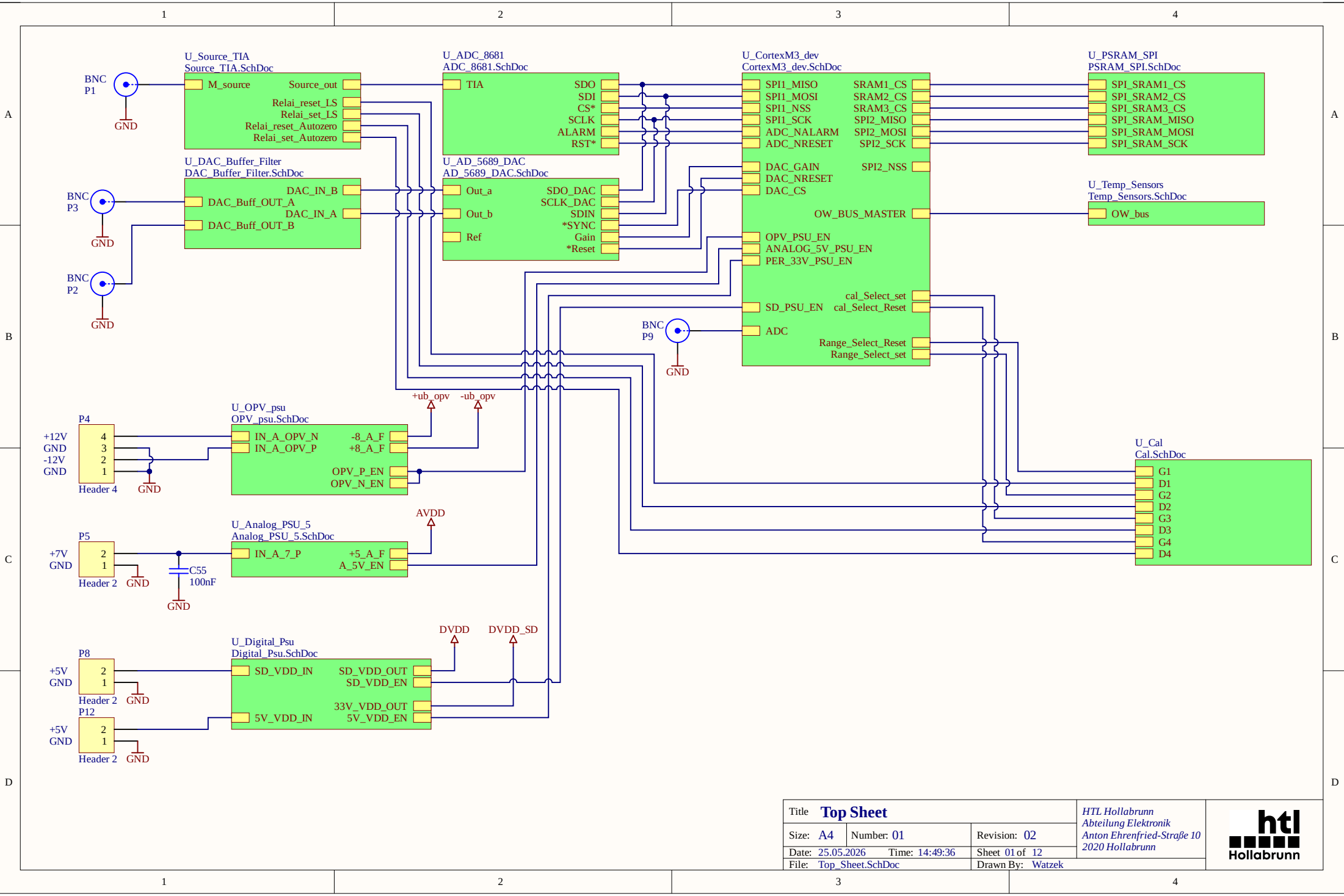
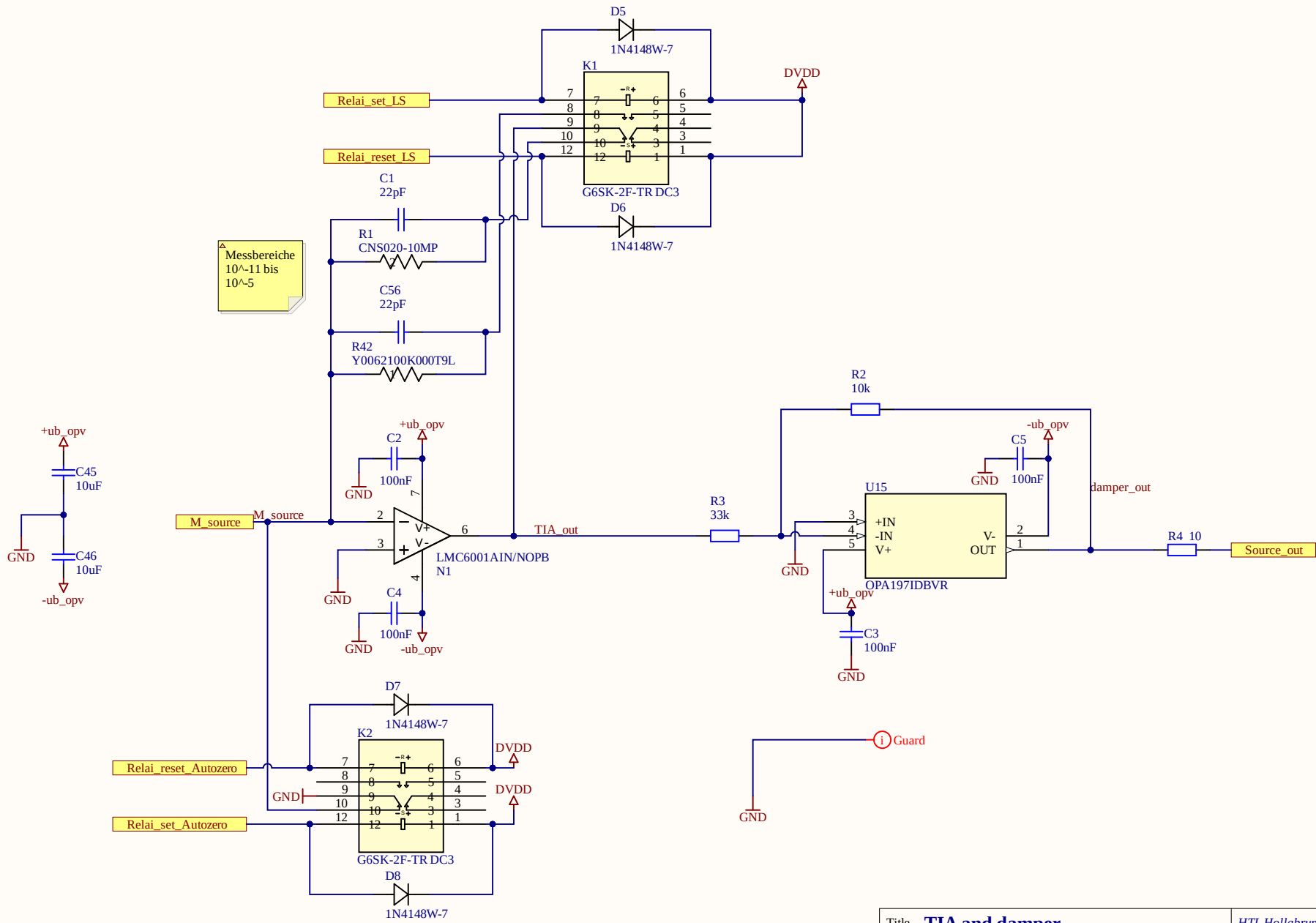


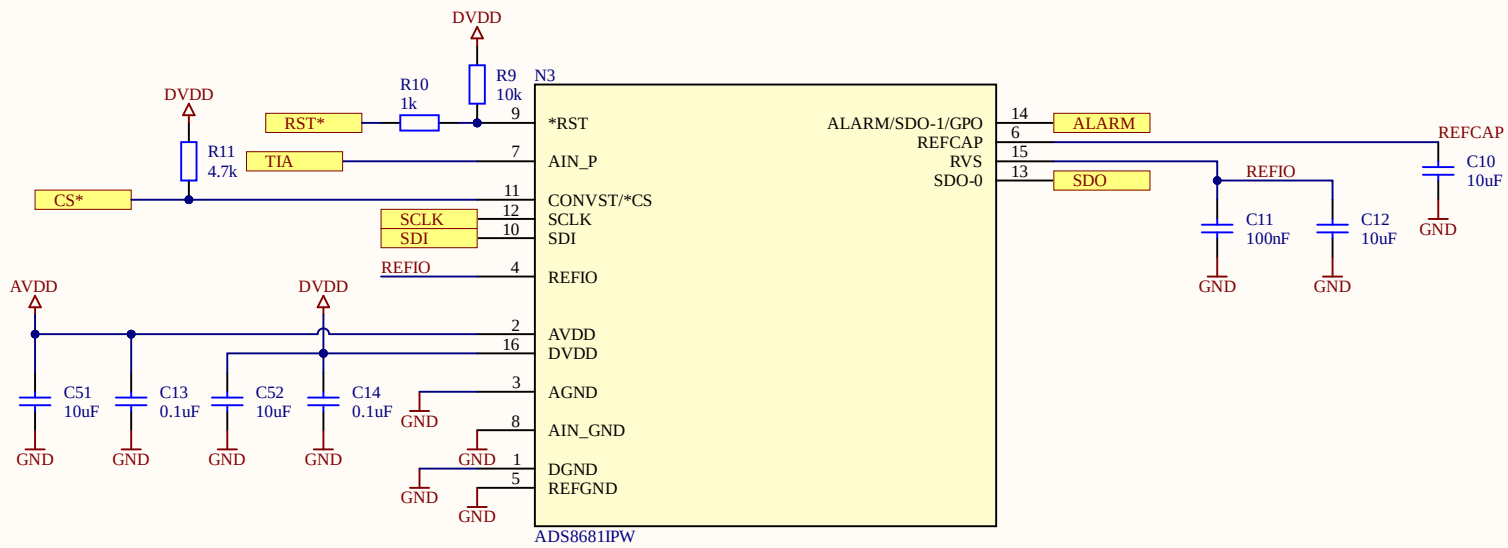


Title Top Sheet		HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn	
Size: A4	Number: 01	Revision: 02	
Date: 25.05.2026	Time: 14:49:36	Sheet 01 of 12	
File: Top_Sheet.SchDoc		Drawn By: Watzek	

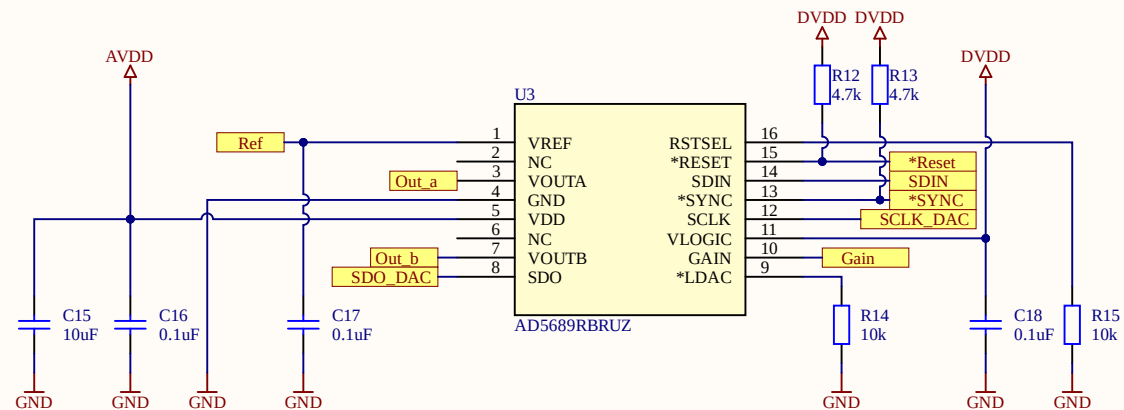




Title TIA and damper			HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn		
Size: A4	Number: 02	Revision: 02			
Date: 25.05.2026	Time: 14:49:37	Sheet 02 of 12			
File: Source_TIA.SchDoc		Drawn By: Watzek			



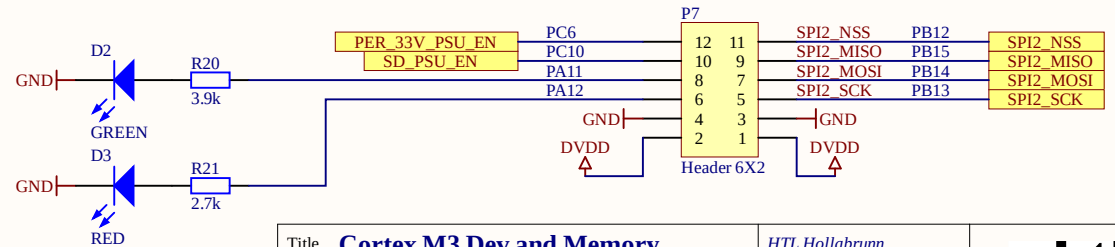
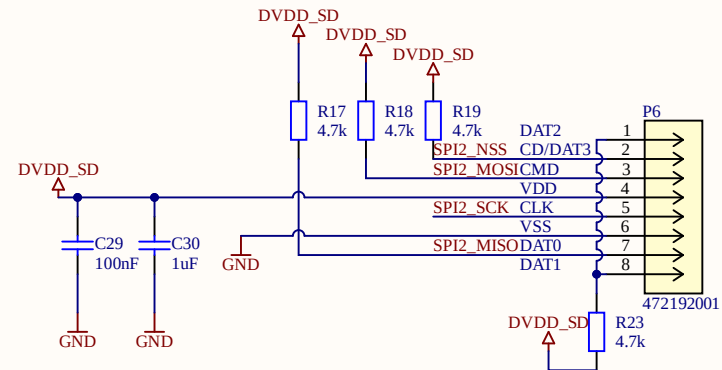
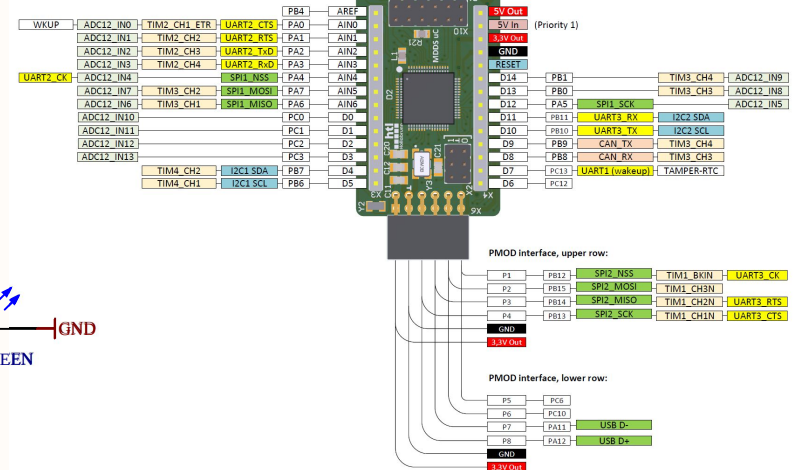
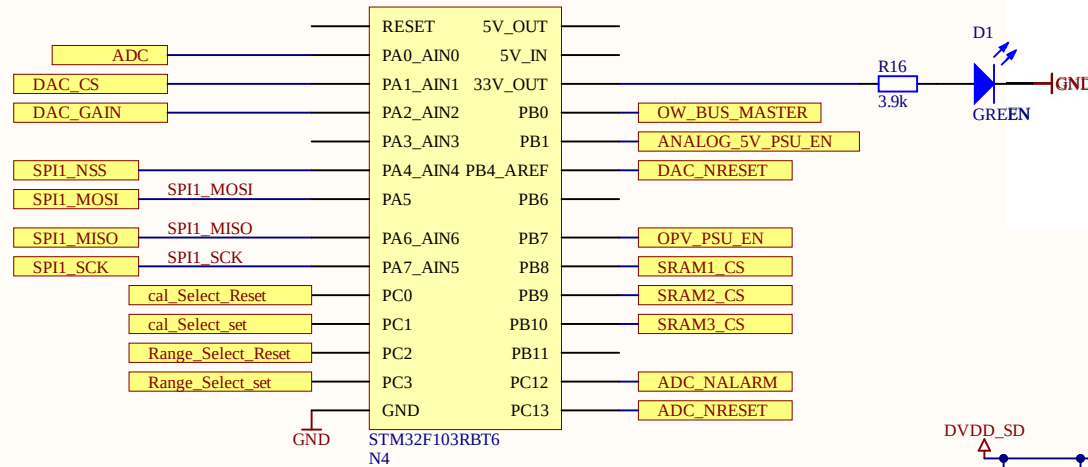
Title source ADC			HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn
Size: A4	Number: 04	Revision: 01	
Date: 25.05.2026	Time: 14:49:37	Sheet 04 of 12	
File: ADC_8681.SchDoc		Drawn By: Watzek	



Title DAC for Gate / Drain			HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn	
Size: A4	Number: 05	Revision: 01		
Date: 25.05.2026	Time: 14:49:37	Sheet 05 of 12		
File: AD_5689_DAC.SchDoc		Drawn By: Watzek		



MDDS_uC (Rev. 9)



GREEN LED for System status is in Order.
RED LED for System has fatal error.
GREEN blinking is for initialisierung

Title **Cortex M3 Dev and Memory**

Size: A4 Number: 06

Date: 25.05.2026 Time: 14:49:37

File: CortexM3_dev.SchDoc

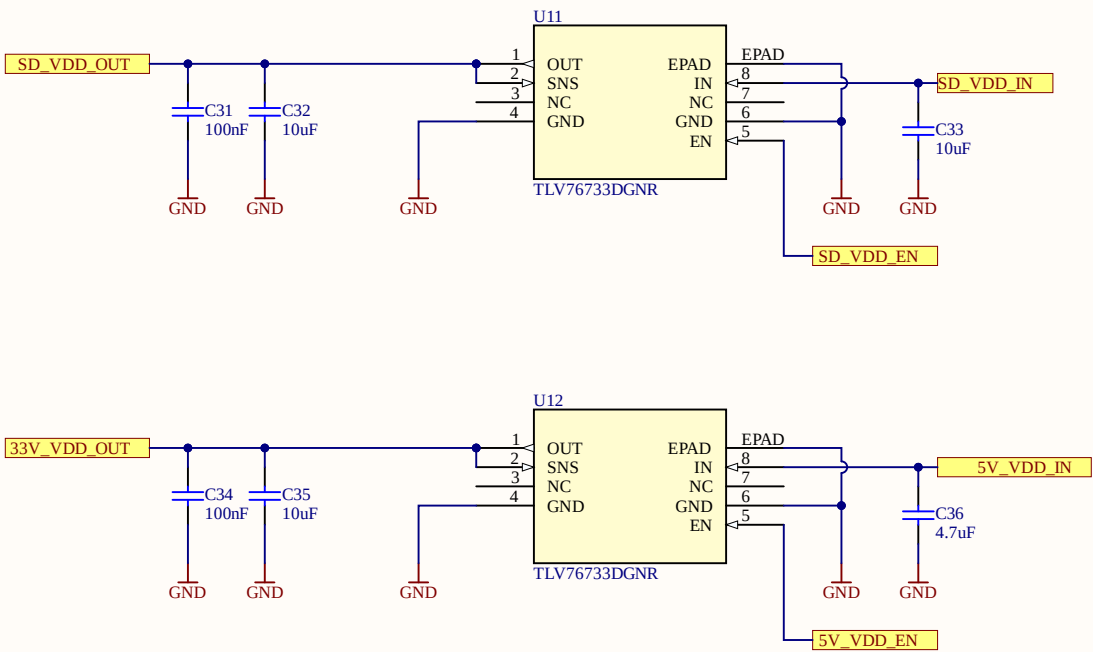
Revision: 01

Sheet 06 of 12

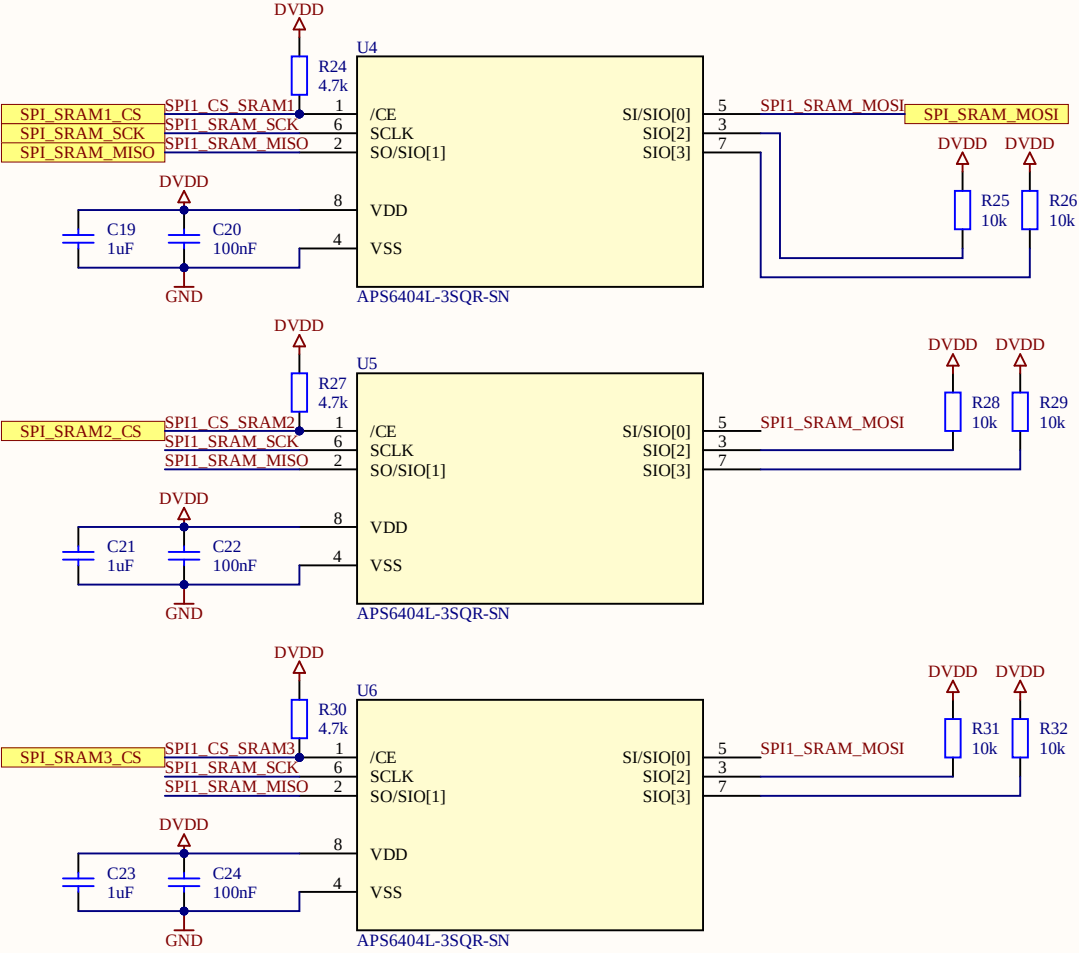
Drawn By: Watzek

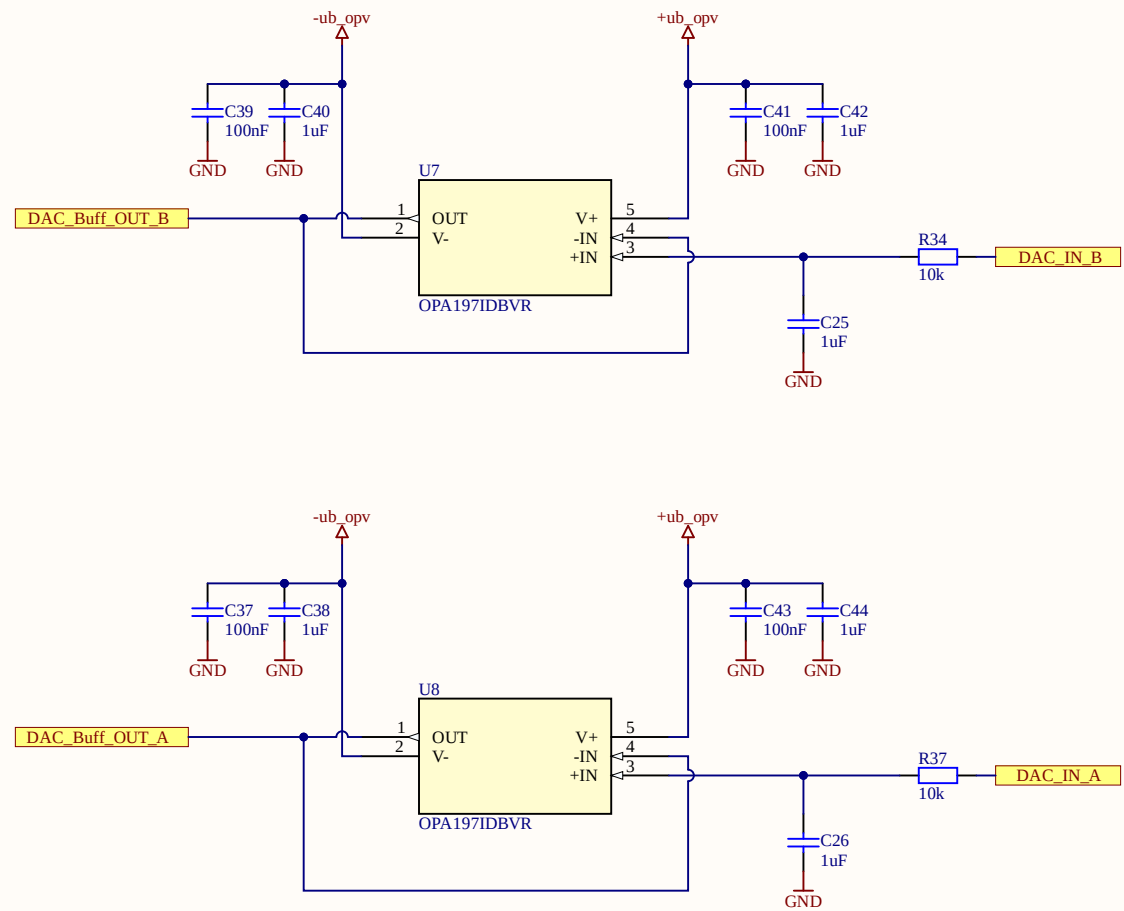
HTL Hollabrunn
Abteilung Elektronik
Anton Ehrenfried-Straße 10
2020 Hollabrunn





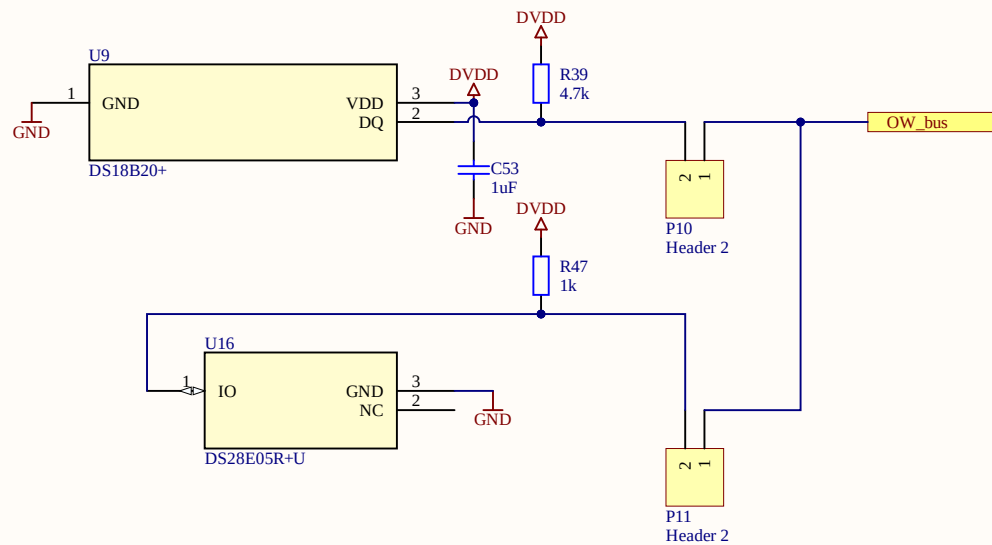
Title Digital PSU			HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn	
Size: A4	Number: 07	Revision: 02		
Date: 25.05.2026	Time: 14:49:37	Sheet 07 of 12		
File: Digital_Psu.SchDoc		Drawn By: Michael Watzek		





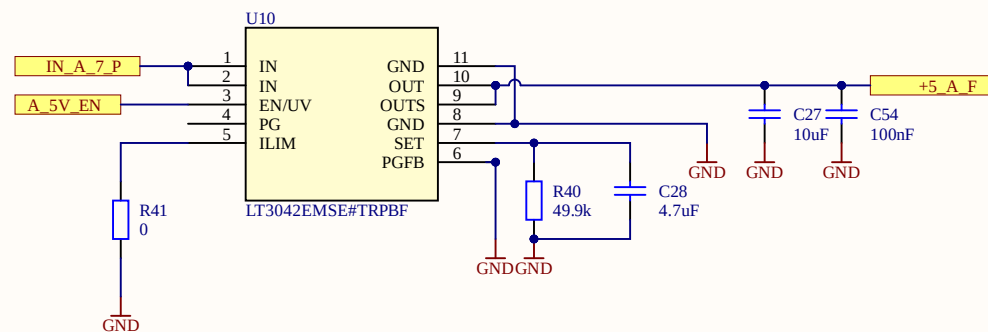
Title DAC Buffer			HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn
Size: A4	Number: 09	Revision: 01	
Date: 25.05.2026	Time: 14:49:38	Sheet 09 of 12	
File: DAC_Buffer_Filter.SchDoc		Drawn By: Michael Watzek	





Title One Wire Devices			HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn	
Size: A4	Number: 10	Revision: 01		
Date: 25.05.2026	Time: 14:49:38	Sheet 10 of 12		
File: Temp_Sensors.SchDoc		Drawn By: Michael Watzek		



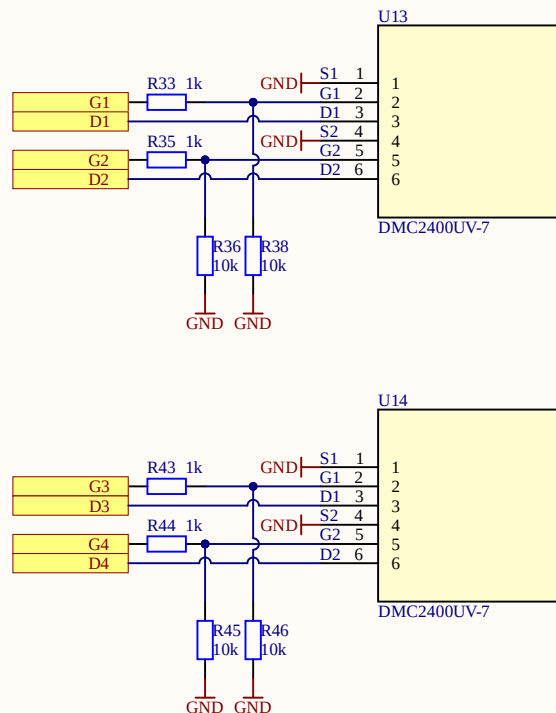


Note that bypass capacitors used to decouple individual components Powered by the LT3094 increase the effective output capacitance.

F output capacitance with an ESR below 10m and an ESL below 1.5nH is required for stability.

Title Analog PSU			HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn	
Size: A4	Number: 11	Revision: 02		
Date: 25.05.2026	Time: 14:49:38	Sheet 11 of 12		
File: Analog_PSU_5.SchDoc		Drawn By: Michael Watzek		





Title Relai MOSFET Controller			HTL Hollabrunn Abteilung Elektronik Anton Ehrenfried-Straße 10 2020 Hollabrunn
Size: A4	Number: 12	Revision: 01	
Date: 25.05.2026	Time: 14:49:38	Sheet 12 of 12	
File: Cal.SchDoc		Drawn By: Michael Watzek	



Comment	Description	Designator	Footprint	LibRef	Quantity
22pF	Capacitor	C1, C56	C1206-KUE	Capacitor	2
100nF	Capacitor	C2, C3, C4, C5, C11, C20, C22, C24, C29, C31, C34, C37, C39, C41, C43, C48, C50, C54, C55	C1206-KUE	Capacitor	19
10uF	Capacitor	C6, C8, C10, C12, C15, C27, C32, C33, C35, C45, C46, C47, C49, C51, C52	C1206-KUE	Capacitor	15
4.7uF	Capacitor	C7, C9, C28, C36	C1206-KUE	Capacitor	4
0.1uF	Capacitor	C13, C14, C16, C17, C18	C1206-KUE	Capacitor	5
1uF	Capacitor	C19, C21, C23, C25, C26, C30, C38, C40, C42, C44, C53	C1206-KUE	Capacitor	11
GREEN	Typical INFRARED GaAs LED	D1, D2	6-0805_N-KUE	LED0	2
RED	Typical INFRARED GaAs LED	D3	6-0805_N-KUE	LED0	1
1N4148W-7	Description	D5, D6, D7, D8	SOD123_DIO	1N4148W-7	4
G69K-2F-TRDC3	Description	K1, K2	RELAY_G69K-2F-DC3_OMR	G69K-2F-TRDC3	2
LMC6001AIN/NOPB	Ultra-Low Input Current Amplifier 8-PDIP-40 to 85	N1	DIP8	LMC6001AIN/NOPB	1
ADS9681IPW	16-bit, 1MSPS 1-ch SAR ADC with programmable ($\pm 1.2V$, $\pm 1.0V$, $\pm 0.6V$, $\pm 0.5V$, $\pm 2.5V$) input ranges on 5V supply 16-TSSOP-40 to 125	N3	PW16_TEX	ADS9681IPW	1
STM32F103		N4	DIP28	STM32F103	1
BNC	BNC Elbow Connector	P1, P2, P3, P9	BNC_RA_CON	BNC	4
Header 4	Header, 4-Pin	P4	HDR4X4	Header 4	1
Header 2	Header, 2-Pin	P5, P6, P10, P11, P12	HDR4X2	Header 2	5
472192001	1.10mm Pitch microSD Card Connector, Hinge Type, Lead-Free	P6	CONN_47219-2001_MOL	472192001	1
Header 6x2	Header, 6-Pin, Dual row	P7	HDR2X6	Header 6x2	1
CNS020-10MP	CNS020 10K0.02%e2	R1	RES_CNS020_VIS	CNS020-10KP	1
10k	Resistor SMD	R2, R9, R14, R15, R25, R26, R28, R29, R31, R32, R34, R36, R37, R38, R45, R46	C1206-KUE	ResistorSMD	16
33k	Resistor SMD	R3	C1206-KUE	ResistorSMD	1
10	Resistor SMD	R4	C1206-KUE	ResistorSMD	1
78.7k	Resistor SMD	R5, R8	C1206-KUE	ResistorSMD	2
0	Resistor SMD	R6, R7, R41	C1206-KUE	ResistorSMD	3
1k	Resistor SMD	R10, R33, R35, R43, R44, R47	C1206-KUE	ResistorSMD	6
4.7k	Resistor SMD	R11, R12, R13, R17, R18, R19, R23, R24, R27, R30, R39	C1206-KUE	ResistorSMD	11
3.9k	Resistor SMD	R16, R20	C1206-KUE	ResistorSMD	2
2.7k	Resistor SMD	R21	C1206-KUE	ResistorSMD	1
49.9k	Resistor SMD	R40	C1206-KUE	ResistorSMD	1
Y0062100K00079L	No Description Available	R42	RES_S102K_7P62X2P6 7_VIS	Y0062100K00079L	1
LT3042EMSETRPBF	20V, 200mA, Ultralow Noise, Ultrahigh PSRR RF-Linear Regulator	U1, U10	05-08-1664_ADI	LT3042EMSETRPBF	2
LT3094EMSETRPBF	$\pm 20V$, 500mA, Ultralow Noise, Ultrahigh PSRR Negative Linear Regulator	U2	MSE_05-08-1666_ADI	LT3094EMSETRPBF	1
AD5689RBRUZ	Dual, 16-Bit nanoDAC+ with 2 ppm/ $^{\circ}$ C Reference, SPI Interface	U3	RU_16_ADI	AD5689RBRUZ	1
APS6404L-3SQR-SN	Description	U4, U5, U6	APS6404L-3SQR-SN_APY	APS6404L-3SQR-SN	3
OPA197IDBVR	Single, 36-V, precision, rail-to-rail input output, low offset voltage op amp 5-SOT-23 -40 to 125	U7, U8, U15	DBV5_TEX	OPA197IDBVR	3
DS18B20+	Programmable Resolution 1-Wire Digital Thermometer	U9	21-0248A_Q8-1_MMM	DS18B20+	1
TLV76733DGNR	Adjustable- and fixed-output, 1-A, 16-V, positive-voltage low-dropout (LDO) linear regulator 8-HVSSOP-40 to 125	U11, U12	VSSOP8_DGN_TEX	TLV76733DGNR	2
DMC2400UV-7	Description	U13, U14	SON_400UV-7_DIO	DMC2400UV-7	2
DS28ED05R+U	1-Wire EEPROM	U16	21-0051H_MMM	DS28ED05R+U	1